A Stochastic Extension of Temperature-Index Snowmelt Modeling for Probabilistic Runoff Estimation in a Himalayan Basin

Avinash Parla

Abstract

Snowmelt-driven hydrological systems in high mountain regions exhibit significant variability due to complex terrain, spatial heterogeneity, and sensitivity to temperature fluctuations. Traditional temperature-index snowmelt models provide deterministic estimates that fail to capture intrinsic variability and extreme melt events.

In this study, we develop a stochastic extension of the classical temperature-index model by incorporating state-dependent variability, temporal persistence through an autoregressive process, and temperature-triggered jump dynamics. The framework is applied to the Chandra Basin in the western Himalayas using ERA5 reanalysis temperature data and DEM-derived basin representation.

Monte Carlo simulations are used to generate probabilistic estimates of snowmelt and runoff. Results show that while mean melt behavior remains consistent with deterministic estimates, stochastic formulations significantly enhance variability and produce heavy-tailed distributions, indicating increased likelihood of extreme melt events.

The proposed approach provides a computationally efficient pathway for transitioning from deterministic snowmelt estimation to probabilistic hydrological analysis, with implications for risk assessment, water resource management, and climate change studies in snow-dominated regions.

Keywords: Snowmelt modeling, Temperature-index model, Stochastic processes, Monte Carlo simulation, Himalayan hydrology, Runoff uncertainty

1 Introduction

Snow- and glacier-fed river systems play a critical role in sustaining water resources in high mountain regions, particularly across the Himalayas, where seasonal snowmelt contributes substantially to river discharge and downstream water availability [Bookhagen and Burbank, 2010, Immerzeel, 2010]. In many Himalayan basins, melt-driven flow dominates pre-monsoon hydrology, making accurate representation of snowmelt processes essential for understanding water availability and variability [Singh and Jain, 2002, Barnett, 2005].

Snowmelt has traditionally been modeled using deterministic approaches, most commonly through temperature-index (degree-day) formulations. These models establish a direct relationship between air temperature and melt rates and have been widely adopted due to their simplicity, robustness, and minimal data requirements [Hock, 2003, Ohmura, 2001, Rango and Martinec, 1995]. More physically detailed energy balance models explicitly account for radiative and turbulent fluxes governing melt processes, offering improved physical realism at the expense of increased data demands and computational complexity [Marks, 1999, Essery, 2013]. Conceptual hydrological models such as HBV and SRM further integrate snowmelt into basin-scale runoff simulations, providing useful tools for operational forecasting and water resource assessment [Bergström, 1976, Lindström, 1997, Martinec, 2008].

Despite their widespread use, deterministic snowmelt models provide single-valued estimates of melt and runoff, implicitly assuming that the governing processes can be fully described through known relationships and parameters. In reality, snowmelt processes are inherently variable, influenced by spatial heterogeneity in snow distribution, elevation-dependent temperature gradients, and localized meteorological conditions [Dozier, 2011, Essery, 2013]. These factors introduce variability that cannot be fully captured within deterministic frameworks, particularly in complex mountainous terrain.

To address uncertainty in hydrological systems, various stochastic and probabilistic approaches have been developed. These include ensemble modeling, stochastic perturbation of meteorological inputs, and probabilistic parameter estimation [Montanari, 2007, Beven, 2001]. Monte Carlo methods are commonly used to propagate uncertainty through hydrological models, enabling estimation of output distributions and extreme events [Koutsoyiannis, 2003, Robert and Casella, 2004]. However, in most existing approaches, uncertainty is introduced externally through inputs or parameters, while the underlying snowmelt model itself remains deterministic.

This separation between deterministic process representation and externally imposed uncertainty limits the ability of models to capture intrinsic variability arising from unresolved physical processes. In snow-dominated systems, variability is often strongly dependent on environmental conditions, particularly temperature, where small perturbations can lead to significant differences in melt rates during active ablation periods.

In this study, we propose a stochastic extension of the classical temperature-index snowmelt model in which variability is incorporated directly into the melt formulation. The approach represents melt as a combination of a deterministic component and a state-dependent stochastic term, allowing the model to generate a distribution of possible outcomes rather than a single estimate. Temporal persistence in variability is represented using an autoregressive process, while intermittent extreme melt events are captured through a temperature-triggered stochastic component.

The framework is applied to the Chandra Basin in the western Himalayas, a high-altitude, snow-dominated catchment where melt processes strongly influence seasonal hydrology. Using reanalysis-based temperature data and DEM-derived basin representation, the study aims to (i) implement a baseline deterministic snowmelt model, (ii) develop a stochastic formulation that captures condition-dependent variability, and (iii) quantify uncertainty in melt and runoff using Monte Carlo simulation.

By integrating deterministic modeling with internally generated stochastic variability, this work provides a pathway for transitioning from traditional snowmelt estimation toward probabilistic characterization of melt and runoff processes. Such an approach is particularly relevant for high mountain regions, where variability and extremes play a critical role in hydrological response and water resource management.

2 Study Area

The study is conducted in the Chandra Basin, located in the western Himalayas within the Lahaul–Spiti region of Himachal Pradesh, India. The basin forms the upper catchment of the Chandra River, which originates from glaciated terrain near the Bara Shigri Glacier and later joins the Bhaga River to form the Chenab River system.

The basin is characterized by high-altitude mountainous terrain, with elevations ranging from approximately 3,000 m to over 6,000 m above sea level. This strong elevation gradient results in significant spatial variability in temperature, snow accumulation, and melt processes. The region experiences cold winters dominated by snowfall associated with western disturbances, leading to

seasonal snowpack development.

During the pre-monsoon and early summer periods, rising temperatures drive snowmelt, which constitutes a major component of basin runoff. In contrast to monsoon-dominated regions, precipitation during summer is relatively limited, making melt processes the primary driver of hydrological response in the basin.

The terrain is highly complex, with steep slopes, varying aspects, and heterogeneous surface conditions, all of which influence the distribution and persistence of snow. Elevation-dependent temperature gradients and differential solar radiation further contribute to variability in melt rates across the basin.

Hydrologically, the basin is dominated by snow- and glacier-fed processes, with meltwater contributing significantly to streamflow in the Chandra River, a major tributary of the Chenab River system. The steep gradients, combined with pronounced elevation variability, lead to rapid runoff response and complex flow routing mechanisms.

For the purposes of this study, the basin is represented at an aggregated scale using a DEM-derived boundary and reanalysis-based meteorological forcing. While this approach does not explicitly resolve spatial heterogeneity, it allows for clear analysis of basin-scale melt dynamics and the development of a stochastic modeling framework.

A digital elevation model (DEM) derived from Shuttle Radar Topography Mission (SRTM) data was used for basin delineation and terrain analysis. The DEM provides sufficient spatial resolution to capture key geomorphological features influencing snow accumulation and melt processes. The spatial extent, elevation distribution, and terrain characteristics of the basin are shown in Figure 1.

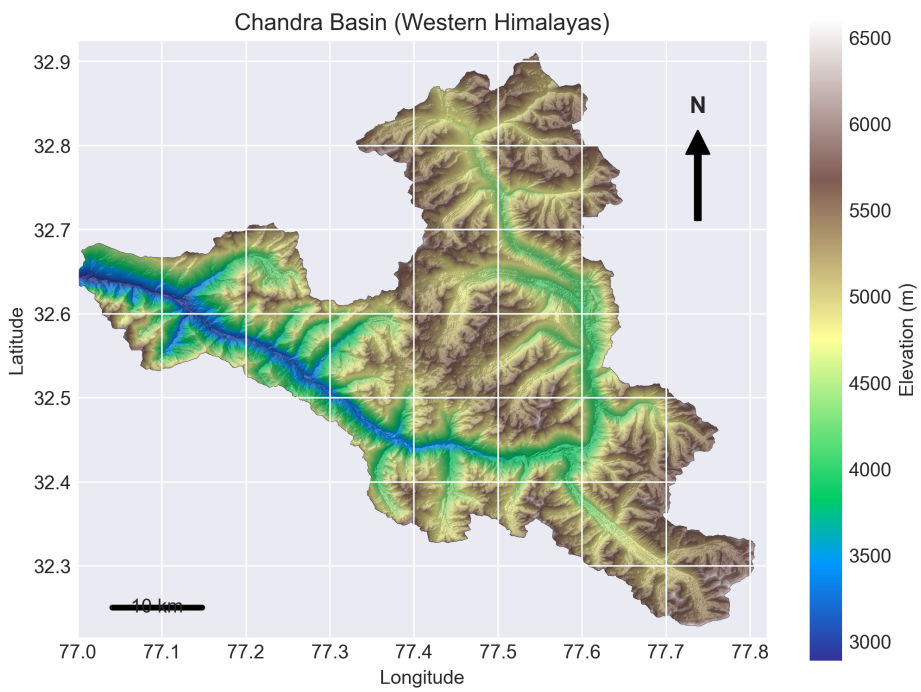

Figure 1: Topographic map of the Chandra Basin in the western Himalayas derived from SRTM DEM. Elevation is shown in meters above sea level with hillshade applied to enhance terrain features. The delineated basin boundary is overlaid in black, along with a north arrow and scale bar for spatial reference.

3 Data

This study integrates reanalysis and topographic datasets to represent meteorological forcing and basin characteristics for the Chandra Basin. The selected datasets provide spatially consistent coverage and are suitable for high-altitude hydrological analysis.

3.1 Meteorological Data

Meteorological forcing is derived from the ERA5 reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) [Hersbach, 2020]. ERA5 provides globally consistent atmospheric variables by assimilating observations into a numerical weather prediction system.

In this study, daily near-surface air temperature (2 m temperature) is used as the primary driver of snowmelt. The ERA5 data are extracted over the spatial extent of the basin and processed to obtain basin-averaged temperature time series. Sub-daily data are aggregated to daily means to match the temporal resolution of the snowmelt model.

Temperature serves as the key input to the temperature-index formulation, capturing seasonal variations that drive snow accumulation and melt processes.

3.2 Topographic Data and Basin Delineation

Topographic characteristics of the basin are derived from the Shuttle Radar Topography Mission (SRTM) digital elevation model [Farr, 2007]. The DEM is used to delineate the basin boundary and generate a spatial mask for extracting basin-specific meteorological data.

The delineation process involves hydrological conditioning of the DEM, including flow direction and flow accumulation analysis, followed by identification of the basin outlet and upstream contributing area. The resulting basin mask is then reprojected to match the coordinate system of the ERA5 dataset.

This mask is used to spatially aggregate meteorological variables, ensuring that all forcing data correspond strictly to the hydrological extent of the basin.

3.3 Data Processing and Basin Averaging

All datasets are processed to ensure temporal and spatial consistency. The key steps include:

- Reprojection of basin boundaries to match ERA5 grid coordinates
- Construction of a basin mask from DEM-derived polygons
- Extraction of grid points within the basin extent
- Computation of basin-averaged temperature time series

The modeling framework operates at a daily temporal resolution and uses spatially aggregated variables to focus on basin-scale dynamics. While this approach simplifies spatial heterogeneity, it provides a tractable framework for developing and evaluating stochastic extensions of snowmelt modeling.

The analysis period spans multiple years to capture repeated seasonal cycles of snow accumulation and melt, ensuring robust characterization of variability and extremes.

4 Background: Snowmelt Modeling and Uncertainty

Snowmelt modeling plays a central role in hydrological analysis of cold and high-altitude regions, where seasonal snow accumulation and ablation strongly influence runoff generation. Over the past decades, a range of modeling approaches have been developed, broadly categorized into empirical temperature-index models and physically based energy balance models.

Temperature-index models, also known as degree-day models, are among the most widely used approaches due to their simplicity and minimal data requirements. These models assume a linear relationship between air temperature and snowmelt, providing a computationally efficient approximation of melt processes Hock [2003], Ohmura [2001]. Despite their empirical nature, they have been successfully applied in mountainous regions where data availability is limited.

In contrast, energy balance models attempt to explicitly resolve the physical processes governing snowmelt, including radiative fluxes, sensible and latent heat exchange, and ground heat flux Marks [1999], Essery [2013]. While these models offer greater physical realism, they require detailed meteorological inputs and are computationally more demanding, limiting their applicability in data-scarce regions.

Conceptual and semi-distributed hydrological models, such as the HBV model and the Snowmelt Runoff Model (SRM), integrate simplified snowmelt formulations within broader runoff frameworks Bergström [1976], Lindström [1997], Martinec [2008]. These models are widely used for basin-scale simulations but rely on calibrated parameters and simplified process representations.

Despite their widespread use, deterministic snowmelt models provide single-valued estimates of melt and runoff, implicitly assuming that system behavior can be fully described through known relationships and parameters. However, snow and cryospheric systems are inherently variable due to spatial heterogeneity, temporal variability in meteorological forcing, and unresolved sub-grid scale processes Dozier [2011], Essery [2013].

To address these limitations, stochastic and probabilistic approaches have been introduced in hydrological modeling. Common methods include stochastic perturbation of meteorological inputs, ensemble simulations, and probabilistic parameter estimation Montanari [2007], Beven [2001]. Monte Carlo techniques are frequently used to propagate uncertainty and generate distributions of possible outcomes Koutsoyiannis [2003].

However, in most existing approaches, uncertainty is introduced externally through inputs or parameters, while the underlying snowmelt model remains deterministic. This separation limits the ability of models to represent intrinsic variability arising from unresolved physical processes.

In reality, snowmelt variability is often state-dependent, particularly near melting conditions where small perturbations in temperature or energy fluxes can lead to large differences in melt rates. This highlights the need for modeling frameworks that incorporate stochastic variability directly within the governing equations of snowmelt.

The present study addresses this gap by extending the classical temperature-index model through the inclusion of state-dependent stochastic components. By embedding variability within the melt formulation itself, the approach enables probabilistic characterization of snowmelt and runoff, providing a bridge between deterministic modeling and risk-based hydrological analysis.

5 Methodology

The modeling framework adopted in this study builds upon the classical temperature-index (degree-day) snowmelt formulation and extends it by incorporating stochastic variability to represent unresolved physical processes. The methodology consists of four main components: (i) deterministic

snowmelt modeling, (ii) stochastic variability representation, (iii) runoff estimation, and (iv) probabilistic simulation using Monte Carlo methods.

5.1 Deterministic Snowmelt Model

Snowmelt is estimated using a temperature-index approach, which relates melt to air temperature above a threshold value:

$$M_t = \mathrm{DDF} \cdot \max(T_t - T_0, 0) \tag{1}$$

where M_t is the snowmelt at time t, DDF is the degree-day factor (mm °C^{-1} day^{-1}), T_t is basin-averaged air temperature, and T_0 is the threshold temperature, taken as 0°C.

Temperature data are derived from ERA5 reanalysis and aggregated over the basin using a DEM-based spatial mask. The deterministic model provides a baseline estimate of melt driven solely by temperature.

5.2 Stochastic Snowmelt Formulation

To account for variability arising from unresolved processes such as spatial heterogeneity, sub-grid scale energy flux variability, and measurement uncertainty, a stochastic component is incorporated directly into the melt formulation:

$$M_t = M_t^{\mathrm{det}} + \sigma_t \, \epsilon_t \tag{2}$$

where M_t^{det} is the deterministic melt, ϵ_t is a stochastic process, and σ_t is a state-dependent scaling term controlling variability magnitude.

This formulation allows the model to generate a distribution of melt outcomes rather than a single deterministic estimate.

5.3 Autoregressive Representation of Temporal Persistence

Temporal persistence in snowmelt variability is represented using a first-order autoregressive [AR(1)] process:

$$\epsilon_t = \phi \, \epsilon_{t-1} + \eta_t \tag{3}$$

where ϕ is the persistence parameter ($0 < \phi < 1$), and $\eta_t \sim \mathcal{N}(0, 1)$ is Gaussian white noise.

This structure captures memory effects in melt processes, reflecting the influence of antecedent snowpack and atmospheric conditions.

5.4 State-Dependent Variability

The magnitude of stochastic variability is modeled as a function of melt conditions:

$$\sigma_t = a + b \cdot M_t^{\mathrm{det}} \tag{4}$$

where a and b are empirical coefficients.

This reflects the physical expectation that variability is low during cold conditions (no melt) and increases during active melt periods.

5.5 Temperature-Triggered Jump Process

To represent intermittent high-magnitude melt events, a temperature-dependent jump process is introduced. Jump occurrences are defined as:

$$J_t = \begin{cases} Z_t, & \text{if } T_t > T_c \text{ and } U_t < p \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where T_c is a critical temperature threshold, $U_t \sim \mathcal{U}(0,1)$ is a uniform random variable, p is the probability of jump occurrence, and Z_t is a random jump magnitude.

The jump magnitude is modeled as a temperature-dependent stochastic variable, allowing stronger melt pulses under warmer conditions. This component captures episodic melt behavior that cannot be represented by smooth deterministic or autoregressive processes alone.

The final stochastic process becomes:

$$\epsilon_t = \phi\, \epsilon_{t-1} + \eta_t + J_t \tag{6}$$

5.6 Runoff Estimation

Snowmelt is converted to runoff using a linear transformation:

$$Q_t = k \cdot M_t \tag{7}$$

where Q_t is runoff and k is a runoff coefficient representing basin-scale hydrological response.

While simplified, this approach enables consistent comparison between deterministic and stochastic melt dynamics at the basin scale.

5.7 Monte Carlo Simulation Framework

Uncertainty in snowmelt and runoff is quantified using Monte Carlo simulation. Multiple realizations of the stochastic model are generated by sampling the stochastic components:

$$\eta_t \sim \mathcal{N}(0,1) \tag{8}$$

Each realization produces a time series of melt and runoff, resulting in an ensemble of possible outcomes. Statistical measures such as mean behavior, variance, and extreme quantiles (e.g., 90th and 99th percentiles) are derived from the ensemble.

This framework enables transition from deterministic prediction to probabilistic characterization of snowmelt and runoff processes.

5.8 Summary of Model Structure

The proposed framework extends the classical temperature-index model by integrating:

- State-dependent stochastic variability
- Temporal persistence via autoregressive processes
- Temperature-triggered jump dynamics
- Monte Carlo-based uncertainty quantification

By embedding stochasticity directly within the melt formulation, the model captures intrinsic variability in snowmelt processes rather than relying solely on external uncertainty inputs. This provides a more realistic representation of melt dynamics in complex mountainous environments.

6 Results

The proposed stochastic snowmelt framework is evaluated by comparing deterministic and stochastic model outputs, analyzing ensemble-based uncertainty, and examining the statistical characteristics of melt and runoff.

6.1 Temporal Dynamics of Snowmelt

Figure 2 illustrates the temporal evolution of snowmelt simulated using the deterministic model, the AR(1)-based stochastic model, and the temperature-triggered jump model.

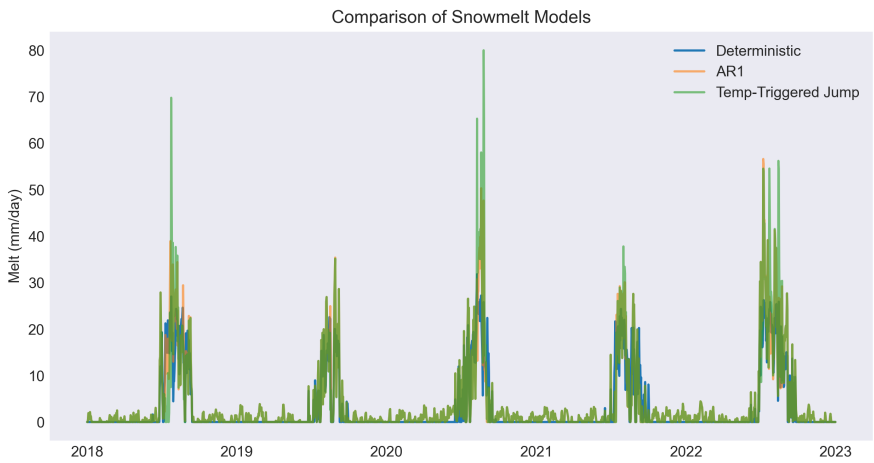

Figure 2: Comparison of deterministic and stochastic snowmelt models. The AR(1) model introduces temporal variability, while the temperature-triggered jump model captures intermittent high-magnitude melt events.

The deterministic model produces smooth seasonal melt cycles driven solely by temperature variations. The AR(1) model introduces temporal persistence, resulting in realistic short-term fluctuations around the deterministic baseline. The temperature-triggered jump model further enhances variability by generating intermittent high-magnitude melt events during warm conditions.

These results demonstrate that stochastic formulations capture both continuous variability and abrupt melt responses, which are not represented in traditional deterministic approaches.

6.2 Uncertainty in Snowmelt Estimates

Uncertainty is quantified using Monte Carlo simulation, generating an ensemble of possible melt realizations.

As shown in Figure 3, uncertainty is strongly state-dependent. During cold periods, melt remains close to zero with minimal variability. In contrast, during active melt periods, the uncertainty band widens significantly, reflecting increased sensitivity to stochastic perturbations.

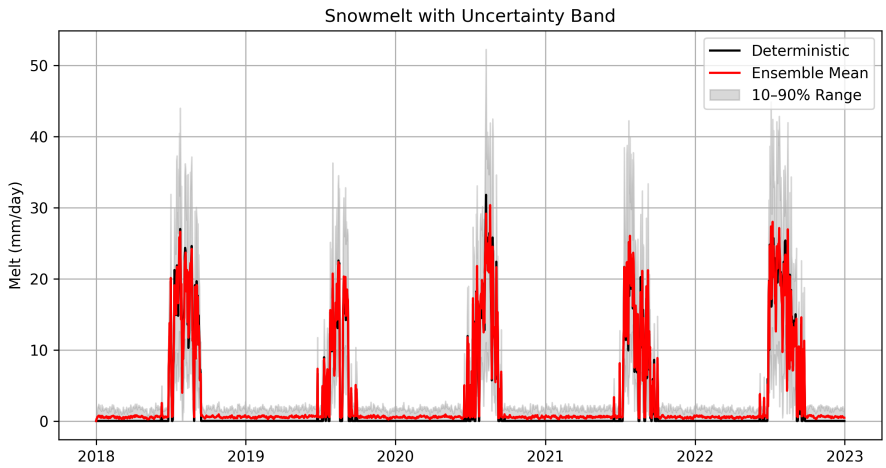

Figure 3: Snowmelt with uncertainty band derived from Monte Carlo simulation. The shaded region represents the 10–90% range, and the ensemble mean represents average stochastic behavior.

The ensemble mean closely follows the deterministic estimate while exhibiting enhanced variability, indicating that stochastic effects primarily influence variability and extremes rather than mean system behavior.

6.3 Distribution of Snowmelt

The distribution of snowmelt values is presented in Figure 4, comparing deterministic and stochastic simulations on a logarithmic density scale.

Both models exhibit high frequency near low melt values; however, the stochastic model produces a substantially broader distribution with a pronounced heavy tail. This indicates an increased probability of moderate and extreme melt events compared to the deterministic formulation.

The heavy tail behavior arises from the combined influence of autoregressive variability and temperature-triggered jumps, which amplify melt responses during favorable thermal conditions.

6.4 Statistical Characteristics of Melt and Runoff

Table 1 summarizes key statistical metrics for melt and runoff across model configurations.

Table 1: Statistical summary of melt and runoff across deterministic and stochastic models.

Model	Mean	Std Dev	Max	P90	P95
Melt - Deterministic	2.69	6.36	31.80	14.16	19.44
Melt - AR1	2.63	7.68	56.60	14.06	22.09
Melt - Temp Jump	3.79	8.72	80.00	13.95	23.36
Runoff - Deterministic	1.96	4.45	22.26	9.91	13.61
Runoff - AR1	2.54	5.37	39.62	9.84	15.41
Runoff - Temp Jump	2.66	6.10	56.00	9.76	16.35

This highlights the importance of incorporating stochastic processes when assessing hydrological extremes, particularly in snow-dominated basins. The stochastic models exhibit substantially

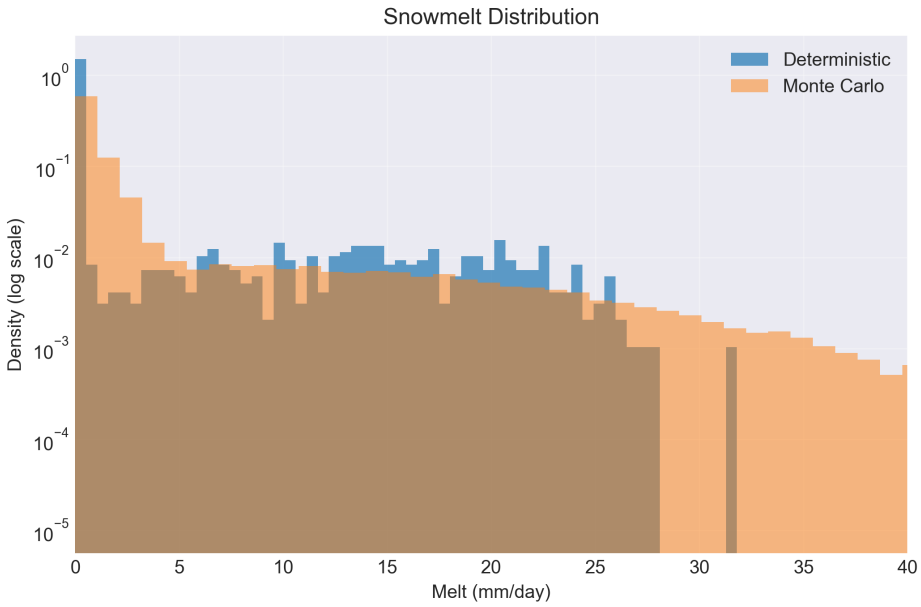

Figure 4: Distribution of snowmelt values on a logarithmic density scale. The stochastic model exhibits a broader distribution with a pronounced heavy tail, indicating increased likelihood of extreme melt events.

higher variability and extreme values compared to the deterministic model. In particular, the temperature-triggered jump model produces the largest maximum melt and runoff values, highlighting its ability to represent extreme melt events.

Despite these differences, mean values remain relatively consistent across models, indicating that stochastic extensions primarily influence variability and extremes rather than average system behavior.

6.5 Summary of Model Behavior

Overall, the results demonstrate that the proposed stochastic framework successfully extends the classical temperature-index model by capturing both variability and extreme melt behavior. The autoregressive component introduces realistic temporal persistence, while the jump process enables representation of episodic high-intensity melt events.

This combined approach provides a more comprehensive representation of snowmelt dynamics in complex mountainous environments.

7 Discussion

The results demonstrate that incorporating stochastic variability directly within the temperature-index snowmelt formulation significantly improves the representation of melt dynamics in a high-altitude Himalayan basin. Unlike deterministic models, which provide a single estimate of melt, the proposed framework captures both temporal variability and the occurrence of extreme melt events.

7.1 Role of Stochastic Variability

The inclusion of an autoregressive stochastic component introduces temporal persistence in melt variability, reflecting the influence of antecedent conditions such as snowpack evolution and atmospheric forcing. This results in more realistic fluctuations in melt rates compared to the smooth behavior produced by deterministic models.

The state-dependent formulation further ensures that variability is physically consistent, with minimal stochastic influence during cold periods and increased variability during active melt conditions. This aligns with the understanding that snowmelt processes are highly sensitive to temperature variations near the melting threshold.

7.2 Representation of Extreme Melt Events

The temperature-triggered jump process plays a key role in capturing intermittent high-magnitude melt events. These events are particularly important in mountainous regions, where short periods of elevated temperature can lead to rapid snowmelt and sharp increases in runoff.

The results show that the jump component significantly increases the upper tail of the melt and runoff distributions, indicating a higher probability of extreme events. Such behavior cannot be represented within traditional deterministic or purely autoregressive frameworks.

7.3 Implications for Hydrological Modeling

The proposed framework provides a transition from deterministic prediction to probabilistic characterization of snowmelt and runoff. This is particularly relevant for hydrological applications involving risk assessment, flood forecasting, and water resource management in snow-dominated basins.

By generating an ensemble of possible outcomes, the model allows for estimation of uncertainty ranges and extreme quantiles, which are critical for decision-making under uncertainty. The ability to represent both mean behavior and variability makes the approach suitable for integration into broader hydrological modeling systems.

7.4 Model Limitations

Despite its advantages, the model has several limitations. The use of a temperature-index formulation does not explicitly account for energy balance components such as radiation, wind, and humidity, which can influence melt processes.

Additionally, the basin is represented at an aggregated scale, which does not capture spatial heterogeneity in elevation, aspect, and snow distribution. This simplification may limit the ability of the model to represent localized melt dynamics.

The stochastic components are parameterized using simplified assumptions, and their calibration remains an area for further investigation. In particular, the selection of parameters governing persistence, variability, and jump behavior can influence model outcomes.

Additionally, the absence of direct observational validation (e.g., satellite-derived snow cover or in-situ measurements) limits the ability to quantitatively assess model accuracy, and future work should address this through integration of observational datasets.

8 Future Work

While the present study focuses on the development and evaluation of a stochastic snowmelt modeling framework, several extensions can further enhance its applicability and physical realism.

A key direction for future work is the integration of satellite-based snow cover observations, such as Moderate Resolution Imaging Spectroradiometer (MODIS) products. Incorporating snow cover data would enable direct validation of modeled melt dynamics through comparison with observed snow depletion patterns. This would provide an important link between modeled processes and real-world observations, strengthening confidence in the stochastic framework.

Future studies can also extend the model to a spatially distributed framework, allowing explicit representation of elevation-dependent processes, aspect variability, and heterogeneous snow distribution. Such an approach would improve the ability to capture localized melt dynamics in complex mountainous terrain.

In addition, coupling the stochastic snowmelt model with hydrological routing models would enable simulation of streamflow response, facilitating applications in flood forecasting and water resource management.

The probabilistic nature of the proposed framework makes it particularly suitable for climate change impact studies. As temperature variability and extreme events are expected to increase under future climate scenarios, the ability to quantify uncertainty and extremes becomes increasingly important.

Overall, these extensions would transform the present framework from a basin-scale conceptual model into a comprehensive tool for probabilistic hydrological analysis in snow-dominated regions.

9 Conclusion

This study presents a stochastic extension of the classical temperature-index snowmelt model for probabilistic estimation of melt and runoff in a high-altitude Himalayan basin. By integrating deterministic modeling with state-dependent stochastic variability, the proposed framework enables representation of both average behavior and uncertainty in snowmelt processes.

The results demonstrate that deterministic models, while effective in capturing general seasonal trends, are limited in their ability to represent variability and extreme events. The inclusion of autoregressive persistence and temperature-triggered jump processes allows the model to capture temporal variability and intermittent high-magnitude melt events that are commonly observed in mountainous environments.

The stochastic formulation preserves the simplicity and computational efficiency of the temperature-index approach while significantly enhancing its ability to represent real-world system behavior. Importantly, the model produces probabilistic outputs in the form of ensembles, enabling estimation of uncertainty ranges and extreme quantiles, which are critical for hydrological risk assessment.

The findings highlight the importance of explicitly representing intrinsic variability in snowmelt-driven systems, particularly in regions where small changes in temperature can lead to significant hydrological responses. The proposed framework provides a conceptual bridge between traditional deterministic snowmelt modeling and probabilistic hydrological analysis.

While the current study adopts a basin-aggregated approach, the methodology can be extended to incorporate spatial variability, more physically based melt formulations, and advanced hydrological routing schemes. The framework is also well-suited for applications in climate change impact assessment, where understanding variability and extremes is essential.

Overall, this work contributes toward advancing snowmelt modeling from deterministic prediction to probabilistic understanding, providing a foundation for improved analysis of variability, uncertainty, and extreme events in cryospheric hydrology. This framework provides a scalable foundation for probabilistic cryospheric modeling, enabling improved risk-aware water resource management in data-scarce mountain regions.

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